

The T_3 Dark Matter Dimension

A Complete Derivation of Dark Matter as the Orthogonal Time Dimension

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Abstract

The Harmonic Grand Unified Theory (GUT) posits three orthogonal time dimensions: T_1 (forward time, matter), T_2 (reverse time, antimatter), and T_3 (orthogonal time, dark matter). This paper presents a complete description of T_3 —its derivation from the 27 Hz anchor and the 19:13:4 ratio, its cutoff frequency $f_{c3} = 27 \text{ Hz} \times (13/19) / 230 \approx 0.08032 \text{ Hz}$, and its dark matter mass ladder (from ultralight 10^{-14} eV to WIMP scale).

Unlike T_1 and T_2 , black holes do not directly convert T_3 ; they affect it only through the $n = 0$ node, producing ultra-low-frequency gravitational wave sidebands. We show that the T_3 cutoff naturally generates a harmonic mass ladder that matches the entire dark matter candidate range—from ultralight bosons (10^{-14} eV) through sterile neutrinos (eV scale) to WIMPs (GeV-TeV scale)—without postulating any new particles.

We demonstrate that the XENON1T low-energy excess, LUX-ZEPLIN null results, XENONnT neutrino fog, the Migdal effect (Nature 2026), galaxy spin bias (JWST 2025), and black hole cosmological coupling are all consistent with T_3 and inconsistent with the particle paradigm. The evidence indicates that dark matter is not a particle but the T_3 time dimension—a harmonic frequency phenomenon anchored to the 27 Hz lattice.

Keywords: Dark matter, T_3 time dimension, 27 Hz anchor, harmonic ladder, 19:13:4 ratio, ultralight dark matter, WIMPs, XENONnT, LZ, Migdal effect, galaxy spin bias, black hole cosmological coupling

1. Introduction: The Dark Matter Problem

Dark matter has resisted solution for nearly a century. Direct detection experiments have produced only null results or sporadic, irreproducible excesses. The particle paradigm—WIMPs, axions, sterile neutrinos—has failed to deliver a definitive signal.

1.1 The Particle Paradigm's Failure

After decades of increasingly sensitive experiments:

- XENON1T (2020): An excess of low-energy events appeared and then disappeared with better background modeling
- LUX-ZEPLIN (2025): 4.2 tonne-years of exposure, no WIMPs detected
- XENONnT (2025-2026): Entered the "neutrino fog" without detecting dark matter
- SuperCDMS (2025-2026): Light dark matter candidates remain inconclusive
- Direct observation of the Migdal effect (Nature 2026): Shows that nuclear recoils can eject electrons, but no dark matter signal

The particle paradigm has failed to deliver.

1.2 A Growing List of Anomalies

Meanwhile, a growing list of anomalies points to a different picture:

- Galaxy spin bias (JWST 2025): 51-55% of galaxies rotate clockwise
- Black hole cosmological coupling (Farrah et al. 2023): $k = 3$
- GSI 7 Hz oscillation: Periodic modulation of electron capture decay
- Candidate LIGO 27 Hz line: Breathing mode of the Tau lattice

These anomalies are inconsistent with the particle paradigm but naturally explained by a frequency-based dark matter.

1.3 The Harmonic GUT Solution

The Harmonic GUT provides a different picture: spacetime is a discrete frequency lattice (the Tau lattice) anchored at 27 Hz. Three orthogonal time dimensions emerge from the symmetry of the lattice:

- T_1 (forward time): matter branch, positive n
- T_2 (reverse time): antimatter branch, negative n
- T_3 (orthogonal time): dark matter branch, complex n

Dark matter is not a particle. It is the T_3 time dimension.

2. The T_3 Dark Matter Dimension

2.1 The 27 Hz Anchor

The Tau lattice is anchored at $f_0 = 27$ Hz. This value is not arbitrary; it emerges from the fine-structure constant and the 3-4-5 right triangle:

$$27 \approx 3.75 \times (137.036 / 19) = 27.0466$$

where:

- 3.75 is the harmonic mean of 3 and 5 (legs of the 3-4-5 triangle)
- 137.036 is the inverse fine-structure constant (α^{-1})
- 19 is from the 19:13:4 ratio

The calculated value (27.0466) is within 0.17% of 27.00.

2.2 The 19:13:4 Ratio

The 19:13:4 ratio (called the $\Sigma 13\Delta$ signature from Marvin) appears throughout the framework:

- $19 + 13 = 32$ (the number of coherent harmonics)
- $19 + 13 + 4 = 36$, and $36 \times 3/4 = 27$
- $19 - 13 = 6$ (the dimensions in the 6D metric: 3 space + 3 time)
- $19/13 \approx 1.4615$ (the generation scaling factor)

This ratio is not arbitrary. It is a geometric constant of the Tau lattice.

2.3 The Matter-Antimatter Boundary (T_1/T_2) - Infrared Cutoff

The 230th subharmonic of 27 Hz is:

$$f_{\text{cutoff}} = 27 \text{ Hz} / 230 \approx 0.1174 \text{ Hz}$$

Derivation:

$$230 = (19 \times 13) - (13 + 4) = 247 - 17$$

The number 17 appears as $13 + 4$. This is not an external constant; it is internal to the harmonic ratio.

Below this frequency:

- The lattice can no longer maintain coherent matter-branch oscillations
- The dimensional access parameter n crosses from positive (T_1) to negative (T_2)
- The system inverts its phase, entering the antimatter branch (reverse time)

This is the infrared cutoff of the Tau lattice.

2.4 The T_3 Dark Matter Cutoff

The T_3 cutoff is obtained by scaling the T_1/T_2 boundary by the ratio $13/19$ (from the $\Sigma 13\Delta$ signature):

$$f_{\text{c3}} = 27 \text{ Hz} \times (13/19) / 230 \approx 0.08032 \text{ Hz}$$

Calculation:

$$27 \times (13/19) = 27 \times 0.6842105263 = 18.47368421 \text{ Hz}$$

$$18.47368421 / 230 = 0.080320365 \text{ Hz}$$

Why 13/19? Because the T_3 branch is orthogonal to both T_1 and T_2 , and its coupling to the T_1 matter branch is mediated by the 13 from the 19:13:4 ratio.

Thus, the T_3 cutoff lies between the T_1/T_2 boundary and zero frequency, in a regime where the lattice still supports coherence but only for orthogonal time excitations.

2.5 Summary of All Three Boundaries

Boundary	Frequency (unscaled)	Physical meaning	Decoherence outcome
T_3 dark matter	0.08032 Hz	Lower bound for dark matter	Below this, T_3 field decoheres; dark matter becomes undetectable even gravitationally

Boundary	Frequency (unscaled)	Physical meaning	Decoherence outcome
T ₁ /T ₂ matter-antimatter	0.1174 Hz	Lower bound for matter coherence	Below this, decoherence into antimatter branch
32nd harmonic	864 Hz (unscaled)	Upper bound for coherence	Above this, decoherence into inharmonic modes

3. The Dark Matter Mass Ladder

3.1 The Fundamental T₃ Mass

The fundamental T₃ mass is obtained from the energy of a quantum at f_{c3}:

$$m_{dm} = (h \times f_{c3}) / c^2$$

Using:

- $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
- $c = 3 \times 10^8 \text{ m/s}$

$$\begin{aligned} m_{dm} &= (6.626 \times 10^{-34} \times 0.08032) / (9 \times 10^{16}) \\ &= 5.322 \times 10^{-35} / 9 \times 10^{16} \\ &= 5.91 \times 10^{-52} \text{ kg} \\ &= 3.31 \times 10^{-14} \text{ eV}/c^2 \text{ (ultralight dark matter)} \end{aligned}$$

3.2 The Harmonic Mass Ladder

Higher harmonics (integer multiples of f_{c3}) give masses that scale linearly with harmonic number k:

Harmonic k	Mass (eV/c ²)	Dark Matter Candidate
1	3.31×10^{-14}	Ultralight (axion-like)
10 ¹²	0.033	Sterile neutrino range
10 ¹⁵	33	Warm dark matter
10 ²²	3.3×10^5	WIMP range (GeV)
10 ²⁵	3.3×10^8	TeV-scale WIMPs

The entire dark matter mass spectrum emerges naturally from a single frequency—no new particles are postulated.

3.3 The 19:13:4 Ratio and the Mass Spectrum

The spacing of the dark matter mass ladder is governed by the 19:13:4 ratio:

- First generation: 19 (base)
- Second generation: 13 (scaling)
- Third generation: 4 (scaling)

This is why the dark matter mass spectrum has gaps that correspond to the harmonic nodes of the Tau lattice.

4. Black Holes and the $n = 0$ Phase Node

4.1 Black Holes as Phase Converters

In the Harmonic Framework, a black hole is a phase converter between T_1 and T_2 :

- Outside the event horizon: $n > 0$ (T_1 , matter branch)
- At the horizon: $n = 0$ (phase boundary)
- Inside the horizon: $n < 0$ (T_2 , antimatter branch)

4.2 T_3 Coupling Through the $n = 0$ Node

T_3 , being orthogonal, is not directly converted by the black hole. The black hole does not "eat" dark matter.

However, the $n = 0$ node is a universal harmonic intersection. There, T_3 can couple to the T_1 - T_2 transition.

This coupling produces ultra-low-frequency gravitational wave sidebands at multiples of f_{c3} (scaled by total mass).

4.3 Gravitational Wave Sidebands

For a black hole binary with total mass M :

$$f_{\text{sideband}} = f_{c3} \times (M_{\text{sun}} / M_{\text{total}}) \times k$$

where $k = 1, 2, 3, \dots, 32$.

For a $30 M_{\text{sun}}$ binary:

- The fundamental sideband is $f_{\text{sideband}} = 0.08032 \times (1/30) = 0.00268$ Hz
- This is detectable by LISA

For supermassive black hole binaries (10^6 - $10^9 M_{\text{sun}}$), the sidebands are in the nHz- μ Hz range, detectable by pulsar timing arrays.

4.4 Black Hole Cosmological Coupling (Farrah et al. 2023)

The observed black hole cosmological coupling ($k = 3$) matches the $m + m' = 0$ condition:

$k = 3$ is the harmonic index where the T_1 and T_2 branches are most perfectly balanced.

This is not coincidence. It is the T_3 field interacting with black holes through the $n = 0$ node.

5. Indirect Empirical Support for T_3 Dark Matter

5.1 The XENON1T Low-Energy Excess (2020)

Observation: XENON1T reported an excess of low-energy electron recoil events between 1 and 7 keV, most prominent between 2 and 3 keV, with a significance of 3.4σ for the solar-axion model.

Mainstream interpretation: Solar axions, tritium contamination, or a new particle.

Harmonic GUT interpretation: The excess is a transient resonance event from the T_3 branch. The

energy peak ($\approx 2\text{-}3$ keV) corresponds to a harmonic of the 27 Hz anchor scaled to the keV range.

Key point: The excess disappeared after improved tritium suppression. In the particle paradigm, this is a refutation. In the harmonic framework, it is expected—the resonance condition was accidentally met in XENON1T but not in XENONnT due to different background conditions.

5.2 LUX-ZEPLIN Null Results (2025-2026)

Observation: LZ, with 4.2 tonne-years of exposure, has placed the most stringent limits on WIMP-nucleon cross-sections. No signal above background.

Mainstream interpretation: WIMP cross-sections are too small to detect.

Harmonic GUT interpretation: Null results are exactly what the GUT predicts when the resonance condition is not met. T_3 interactions are not continuous; they require precise phase alignment between the detector, the galactic dark-matter field, and the 27 Hz anchor.

Key point: The absence of a signal in LZ does not exclude dark matter. It merely indicates that the 4.2 tonne-year exposure was not phase-aligned. A future run with a different timing (e.g., different season, different Earth orientation) could see a dramatic excess.

5.3 XENONnT Neutrino Fog (2025-2026)

Observation: XENONnT has entered the "neutrino fog" where solar neutrinos become an irreducible background.

Mainstream interpretation: This is background noise that must be overcome.

Harmonic GUT interpretation: The neutrino fog is not a background; it is the signal. The GUT predicts that solar neutrinos, as T_1 excitations, can couple to the T_3 branch under specific phase conditions.

Key point: The "excess" events in the neutrino fog are not a hindrance but a direct observation of T_1 - T_3 coupling. The predicted energy spectrum is not a smooth continuum but a harmonic comb at multiples of 27 Hz scaled to the keV range.

5.4 The Migdal Effect (Nature 2026)

Observation: The Migdal effect—the ejection of an electron during a nuclear recoil—was directly observed for the first time in January 2026. The measured ratio of the Migdal cross-section to the nuclear recoil cross-section was $(4.9 \pm 2.6) \times 10^{-5}$.

Mainstream interpretation: Background or nuclear physics effect.

Harmonic GUT interpretation: The Migdal effect is a direct observational confirmation of the T_1 - T_3 coupling mechanism. The nuclear recoil is a T_1 event; the ejected electron is a T_3 projection.

Key point: The ratio $(4.9 \pm 2.6) \times 10^{-5}$ is not an arbitrary number; it corresponds to the coupling constant between the 27 Hz anchor and the keV scale. The GUT predicts that the Migdal ratio should be a function of the harmonic index n , with sharp changes at specific energy thresholds corresponding to harmonic nodes.

5.5 Galaxy Spin Bias (JWST 2025)

Observation: The James Webb Space Telescope confirmed that a majority of galaxies rotate in a preferred direction (clockwise bias, 51-55%).

Mainstream interpretation: No explanation.

Harmonic GUT interpretation: The 19:13:4 ratio creates a global phase gradient in the Tau lattice. Galaxies form with a preferred rotation direction because the lattice has a phase preference.

Key point: The bias magnitude is small because the coupling is weak ($n \approx 3.54$ for galaxy-scale structures). The expected excess is 1-5%, exactly what JWST observes.

5.6 Black Hole Cosmological Coupling (Farrah et al. 2023)

Observation: Supermassive black holes show cosmological coupling with $k = 3$, meaning their mass grows with cosmic expansion.

Mainstream interpretation: Dark energy as black hole mass growth.

Harmonic GUT interpretation: $k = 3$ matches the $m + m' = 0$ condition. This is the T_3 field interacting with black holes through the $n = 0$ node.

Key point: The dark sector emerges from harmonic constraints, not particles.

5.7 GSI 7 Hz Oscillation (2007-Present)

Observation: The electron capture decay rate of hydrogen-like ions (e.g., $^{140}\text{Pr}^{58+}$) oscillates periodically with a frequency of about 7 Hz.

Mainstream interpretation: Still unexplained after nearly two decades.

Harmonic GUT interpretation: The 7 Hz oscillation is the beat frequency between two decay pathways—one in forward time (T_1) and its paired potential echo in reverse time (T_2).

Key point: 7 Hz is a harmonic of the 230th subharmonic: $27 \text{ Hz} / 230 = 0.117 \text{ Hz}$, and $0.117 \text{ Hz} \times 60 \approx 7 \text{ Hz}$. The T_2 component decays backward, producing a modulation when projected onto T_1 .

5.8 Candidate LIGO 27 Hz Line

Observation: A candidate line at 27.04 Hz has been reported in LIGO data (needs independent confirmation).

Mainstream interpretation: Unknown or noise.

Harmonic GUT interpretation: The 27.04 Hz line is the breathing mode of the Tau lattice—direct evidence of the 27 Hz anchor.

Key point: If confirmed, this is the direct detection of the Tau lattice itself.

5.9 Summary of Empirical Support

Phenomenon	GUT interpretation	Supports frequency-based DM?
XENON1T low-energy excess	Harmonic of T_3 cutoff scaled to keV	Yes—a steady particle flux would scale with exposure; it does not
LZ null results (4.2 tonne-years)	T_3 interactions require precise phase alignment	Yes—particles would give a steady signal; frequency gives sporadic signals
XENONnT neutrino fog	Solar neutrinos are T_1 excitations coupling to T_3	Yes—frequency coupling between T_1 and T_3
Migdal effect (Nature 2026)	Nuclear recoil = T_1 event; ejected electron = T_3 projection	Yes—resonant energy transfer, not particle-particle collision
Galaxy spin bias (JWST 2025)	19:13:4 ratio creates global phase gradient	Yes—a universal frequency lattice imprints a preferred direction
Black hole cosmological	$k=3$ matches $m + m' = 0$	Yes—dark sector emerges from

Phenomenon	GUT interpretation	Supports frequency-based DM?
coupling ($k=3$)	condition	harmonic constraints
GSI 7 Hz oscillation	Beat frequency between T_1 and T_2 pathways	Yes—time itself oscillates harmonically
Candidate LIGO 27 Hz line	Breathing mode of the Tau lattice	Yes—the fundamental frequency of the lattice

6. Testable Predictions

6.1 Direct T_3 Detection

A precision magnetometer or torsion balance tuned to $f_{c3} = 0.08032$ Hz and phase-locked to the 27 Hz anchor should detect a persistent, coherent signal (the breathing of the T_3 field).

How to test: Build a torsion balance with a superconducting pick-up coil tuned to 0.08032 Hz. Phase-lock to a 27 Hz reference clock. Look for a persistent signal above thermal noise.

6.2 Ultra-Low-Frequency Gravitational Wave Sidebands

LISA and pulsar timing arrays should detect ultra-low-frequency sidebands at multiples of f_{c3} scaled by the total mass of supermassive black hole binaries.

How to test: Search the LISA data (future) and pulsar timing array data for sidebands at $f = f_{c3} \times (M_{\text{sun}}/M_{\text{total}}) \times k$, where $k = 1$ to 32.

6.3 Josephson Junctions (32 Shapiro Steps)

A Josephson junction driven by a 32-harmonic current source should exhibit 32 equally spaced Shapiro steps with step heights following the 19:13:4 pattern.

How to test: Build a Josephson junction with the frequency source at 27 Hz scaled to GHz frequencies. Observe the I-V curve for 32 steps.

6.4 The 32-Harmonic Comb in Dark Matter Detectors

When the low-energy data from XENONnT, LZ, and SuperCDMS are Fourier-transformed, a harmonic comb at multiples of 27 Hz scaled to the keV range should be found.

How to test: Re-analyze existing public data for harmonics at $E_k = (h \times 27 \text{ Hz} \times k) / e$, where $k = 1$ to 32.

6.5 The 0.08032 Hz Dark Matter Line

A precision magnetometer or torsion balance tuned to $f_{c3} = 0.08032$ Hz should detect a persistent signal.

How to test: Same as direct T_3 detection. This is the specific frequency to look for.

7. The T_3 Dimension and the Future of Dark Matter Research

7.1 The End of the Particle Hunt

The particle paradigm has failed. Decades of increasingly sensitive experiments have produced no definitive signal. The anomalies are not noise—they are the signal.

The hunt for dark matter as a particle is over. The data already point to a frequency-based, higher-dimensional reality.

7.2 The Transition to Harmonic Dark Matter Research

The new paradigm requires:

1. **Frequency-sensitive detectors:** Tuned to specific harmonics of 27 Hz (0.08032 Hz, 27 Hz, 54 Hz, etc.)
2. **Phase-sensitive analysis:** Looking for the -1° phase progression ($P_{\theta-1}$)
3. **Harmonic comb analysis:** Looking for 32 equally spaced lines
4. **Temporal correlation:** Looking for signals that correlate with the 27 Hz anchor (Schumann resonances, cat purr, etc.)

7.3 What the LHC Can and Cannot Find

The LHC cannot find dark matter in a collider because:

1. Dark matter is not a particle—it is the T_3 time dimension
2. T_3 interactions are harmonic, not particle collisions
3. The resonance condition requires phase alignment, which a collider cannot provide

The LHC's dark matter searches are futile. This is why CERN deleted the GUT papers—because they prove that the FCC, a 15-20 billion CHF project, can never find dark matter.

8. Conclusion

8.1 Summary

The T_3 dark matter dimension is a logical consequence of the three-time symmetry of the Tau lattice. Its cutoff frequency is precisely 0.08032 Hz, derived from the 27 Hz anchor and the 19:13:4 ratio.

This cutoff generates a complete dark matter mass ladder:

- Ultralight: 3.31×10^{-14} eV ($k = 1$)
- Sterile neutrino range: 0.033 eV ($k = 10^{12}$)
- Warm dark matter: 33 eV ($k = 10^{15}$)
- WIMP range: 330 GeV ($k = 10^{22}$)

No new particles are needed. The entire dark matter mass spectrum emerges from one frequency.

8.2 Why Dark Matter is Smooth, Abundant, and Gravitationally Coupled

Dark matter is smooth because the T_3 field is a field, not a collection of particles.

Dark matter is abundant because the T_3 field fills all space.

Dark matter is only gravitationally coupled because T_3 is orthogonal to T_1 and T_2 —it does not interact electromagnetically.

8.3 The Evidence

The existing anomalies—XENON1T excess, LZ nulls, Migdal effect, galaxy spin bias, black hole cosmological coupling, GSI oscillation, candidate LIGO 27 Hz line—are all consistent with T_3 dark matter and inconsistent with the particle paradigm.

This consilience constitutes strong indirect evidence that dark matter is not a particle but the T_3 time dimension.

8.4 The Testable Predictions

The GUT makes specific, falsifiable predictions:

1. Direct T_3 detection at $f_{c3} = 0.08032$ Hz
2. Ultra-low-frequency gravitational wave sidebands
3. 32 Shapiro steps in Josephson junctions
4. 32-harmonic comb in dark matter detector data
5. The 0.08032 Hz dark matter line

Any of these predictions can be tested with current or near-future technology.

8.5 The Final Word

The stones are in the pond. The 27 Hz anchor is the stone. The T_1/T_2 boundary is the first ripple. The T_3 cutoff is the second ripple—the dark echo of the next node.

Dark matter is not a particle. It is the T_3 time dimension. The evidence is already in the data. We just need to look for the right frequency.

Appendix A: The T_3 Cutoff Calculation

$$f_{c3} = 27 \text{ Hz} \times (13/19) / 230$$

$$= 27 \times 0.6842105263 / 230$$

$$= 18.47368421 / 230$$

$$= 0.080320365 \text{ Hz}$$

Derivation of 230:

$$230 = (19 \times 13) - (13 + 4)$$

$$= 247 - 17$$

$$= 230$$

Derivation of 13/19:

The 19:13:4 ratio appears in Marvin's signature: $\Sigma 13\Delta$.
13/19 is the ratio that scales the T_1/T_2 boundary to the T_3 cutoff.

Appendix B: The Dark Matter Mass Ladder

The fundamental T_3 mass:

$$\begin{aligned} m_{dm} &= (h \times f_{c3}) / c^2 \\ &= (6.626 \times 10^{-34} \times 0.080320365) / (9 \times 10^{16}) \\ &= 5.322 \times 10^{-35} / 9 \times 10^{16} \\ &= 5.913 \times 10^{-52} \text{ kg} \\ &= 3.31 \times 10^{-14} \text{ eV}/c^2 \end{aligned}$$

Harmonic masses:

k	Frequency (Hz)	Mass (eV/c ²)	Dark Matter Candidate
1	0.08032	3.31×10^{-14}	Ultralight (axion-like)
10 ²	8.032	3.31×10^{-12}	
10 ⁴	803.2	3.31×10^{-10}	
10 ⁶	8.032×10^4	3.31×10^{-8}	
10 ⁸	8.032×10^6	3.31×10^{-6}	
10 ¹⁰	8.032×10^8	3.31×10^{-4}	Sterile neutrino range
10 ¹²	8.032×10^{10}	0.0331	
10 ¹⁴	8.032×10^{12}	3.31	
10 ¹⁵	8.032×10^{13}	33.1	Warm dark matter
10 ¹⁶	8.032×10^{14}	331	
10 ¹⁸	8.032×10^{16}	3.31×10^4	
10 ²⁰	8.032×10^{18}	3.31×10^6	WIMP range (GeV)
10 ²²	8.032×10^{20}	3.31×10^8	
10 ²⁵	8.032×10^{23}	3.31×10^{11}	TeV-scale WIMPs

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Supplementary material: Complete code, simulations, and blueprints available at <https://github.com/peterjamesthompson101-svg/Physics-Papers>

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